

Fall detection on the edge

Quality-aware temporal modeling, independent Raspberry Pi inference, and cloud delivery after reconnection.

Yaorong Huang · Jiahao Zhao · Chengyi Xu
Zhonglin Chang · Jiayun Song

Final project presentation · 7 October 2026

20.56_{fps}

Complete local video pipeline
Raspberry Pi 5 + Hailo-10H

11.9_x

Throughput versus CPU ONNX
Same YOLOv8s-pose comparison

Motion provides the context

A single posture can resemble sitting, kneeling, lying down or falling.

We target indoor assisted-living spaces. Caregivers need an event history they can review, including events recorded while the network is unavailable.

01 Observe locally

Extract body joints near the camera.

02 Interpret a short sequence

Relate descent to surrounding motion.

03 Keep the history

Store decisions locally and deliver them later.

Recent research

TCNTE · Yu et al., 2025

Temporal convolutions with Transformer attention for skeleton sequences.

Fall-Mamba · Zhang et al., 2025

Masked temporal modeling with multimodal fusion.

StoneGAT · Chun et al., 2025

Skeleton geometry and confidence-aware graph attention.

Our contribution: combine bidirectional temporal processing with explicit pose quality, then evaluate and deploy the complete edge–cloud system.

Inference stays on the Raspberry Pi

The cloud receives records. It is not needed to compute a fall prediction.

01

Camera / video

USB capture or local file

02

→ Pose extraction

Hailo-10H · YOLOv8s

03

→ Fall classifier

Pi CPU · MaskedBiMamba

04

→ Local records

SQLite + alert state

At the device

One-second windows, 30 time steps, up to five classifications per second. A local page shows annotated frames, current predictions and pending records.

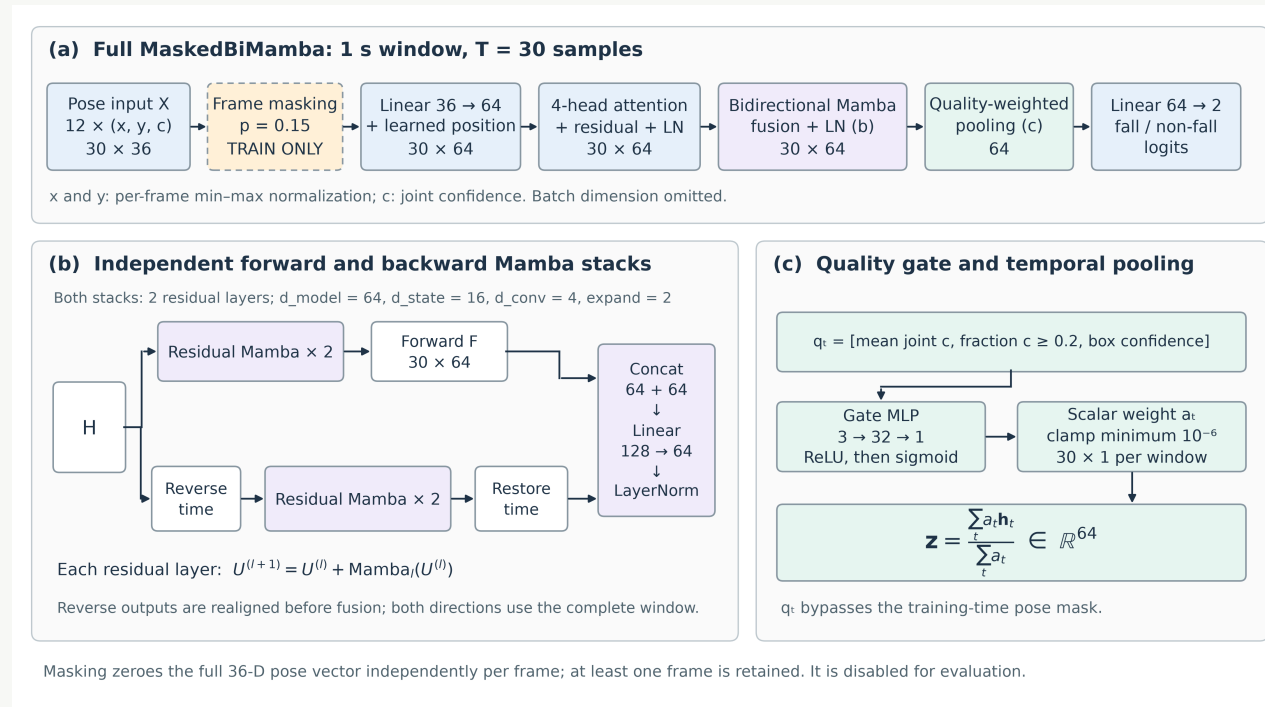
After reconnection

The cloud stores pending records and acknowledges successful writes. Repeated delivery does not create duplicate records, and original capture times are retained.

Implementation: local pose + ONNX temporal classifier; CPU pose inference provides a fallback when Hailo is unavailable.

MaskedBiMamba

Twelve body joints supply normalized coordinates and confidence. A separate stream describes observation quality.



Project architecture. Bidirectional processing uses the complete buffered window; masking is applied during training.

Temporal context

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d}) \mathbf{V}$$

Attention relates positions; forward and reverse Mamba branches model motion in both directions.

Quality-weighted pooling

$$\alpha_t = a_t / \sum_u a_u \quad ; \quad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t$$

A learned gate uses mean joint confidence, visible-joint fraction and person confidence.

Separate protocols, explicit metrics

Data used in the reported evaluation

Dataset	Videos	Role	Window count
CAUCAFall	100	Primary, 10 participants	7,869 combined
GMDCSA-24	160	Primary, 4 subject groups	
Le2i	190	Separate baseline comparison + external test	12,650
OmniFall-Synthetic	40	Edge deployment sample: 20 fall / 20 non-fall	663 device predictions

Primary comparison

Four subject-disjoint folds × three seeds = 12 evaluations per model. Training uses Adam, 100 epochs and validation F1 for model selection. The same windows and pose features are used across five architectures.

What the numbers mean

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

Reported error bars are sample standard deviations, not confidence intervals.

Precision and recall lead to different choices

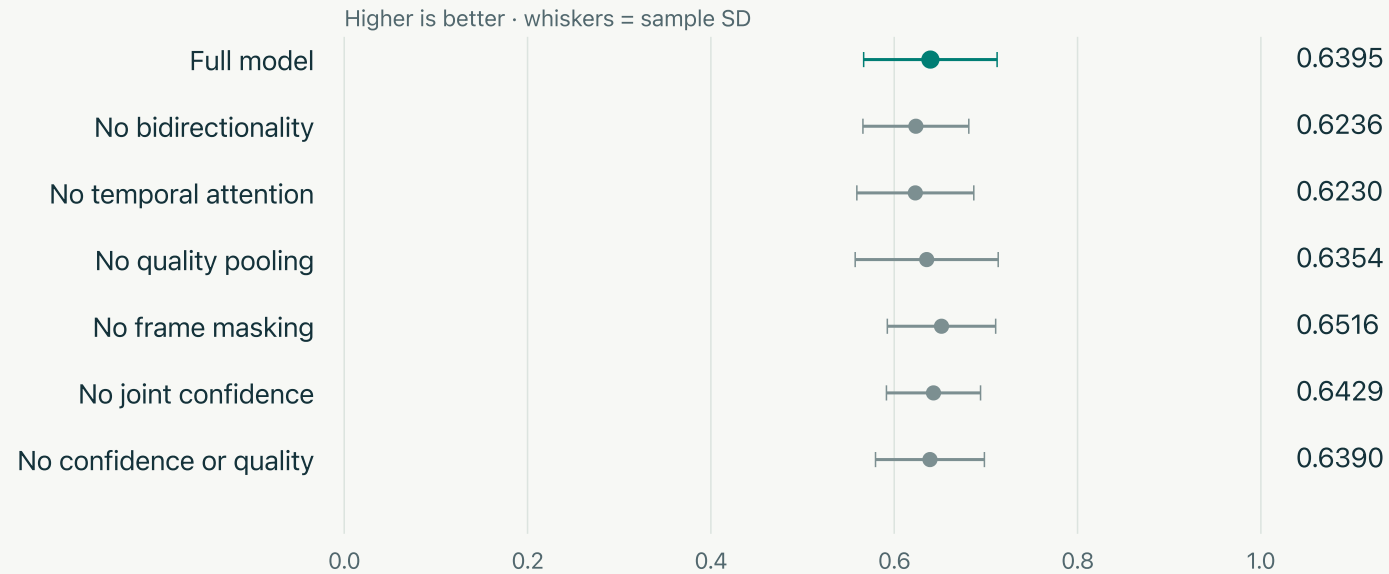


TE has the highest clean window F1: 0.6581. It also has the highest event recall, 0.9767.

MaskedBiMamba has the highest window precision: 0.6048. Its video F1 is 0.7632 and its unmatched-event rate is the lowest, at 198.5 per evaluated hour.

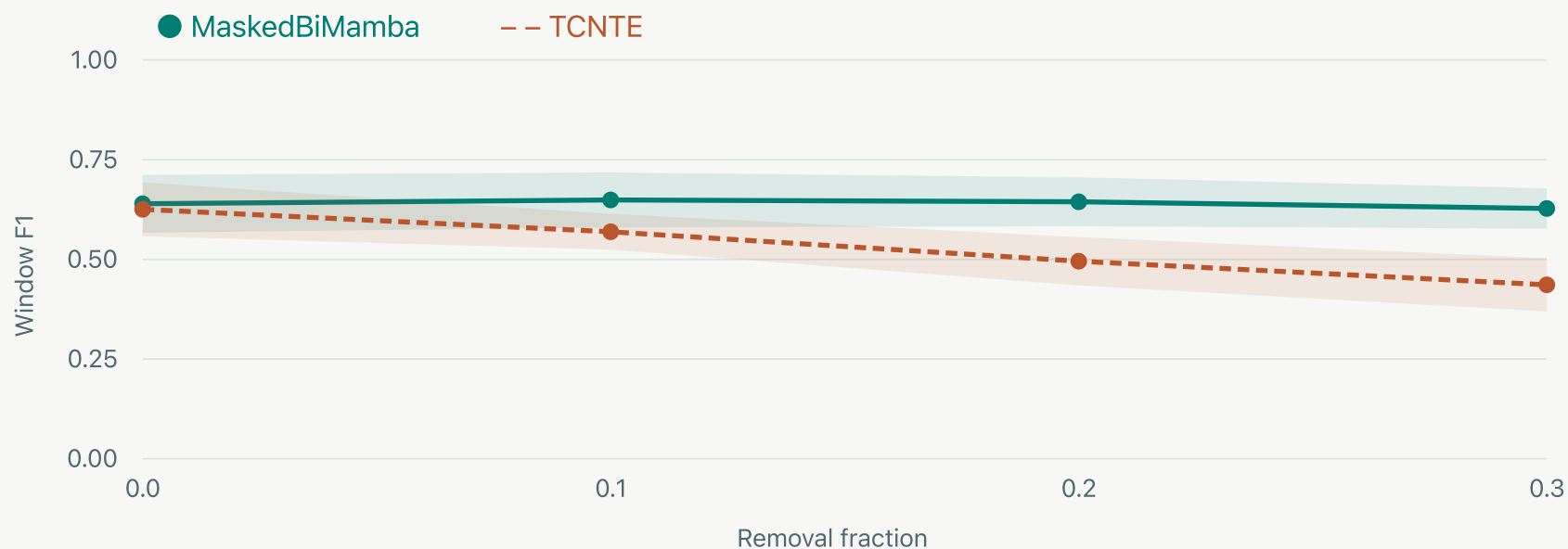
Fixed-rule replay yields confirmed-event recall 0.7752 ± 0.0958 and 66.7 ± 45.2 unmatched confirmed alerts per evaluated hour. These are segmented-video rates, not continuous-care notification rates.

Which changes contribute?



Bidirectional processing and temporal attention contribute to window classification: removing either reduces F1. Masking and quality pooling target incomplete observations; their settings can be calibrated for the deployment conditions.

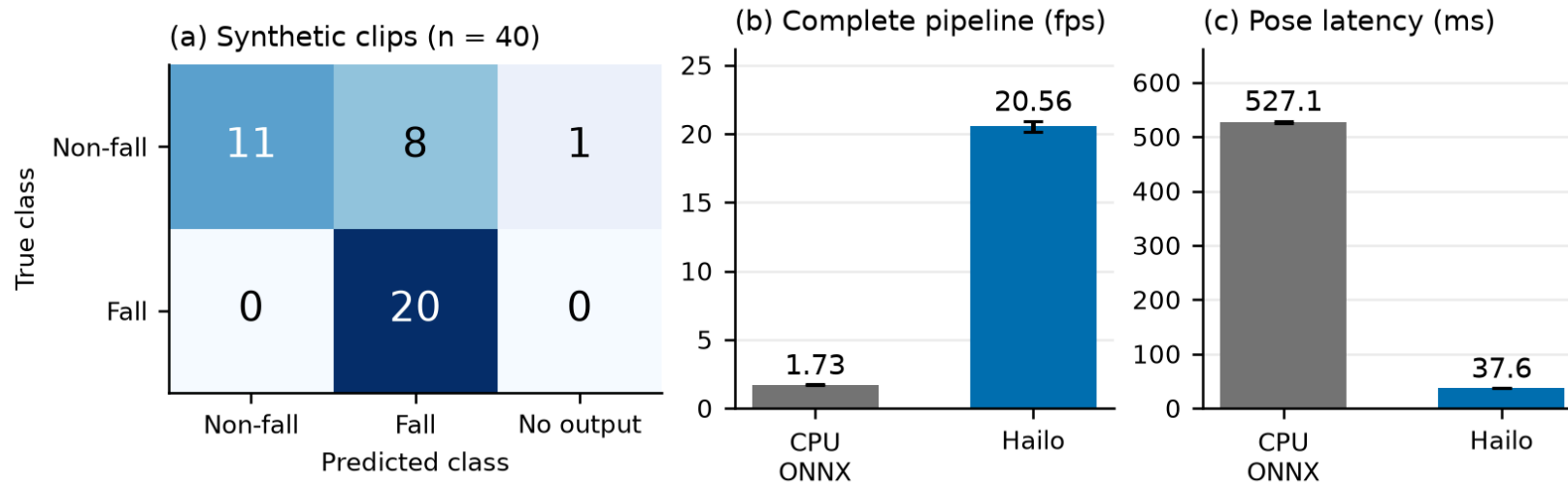
Classification is retained under frame removal



Primary test partitions · Frame removal (0.3): window f1 is 0.6276 for MaskedBiMamba and 0.4361 for TCNTE.

Perturbations are applied to extracted poses. The current comparison covers frame removal and confidence noise; camera and lighting robustness remain untested.

Complete inference without an external network



Same YOLOv8s-pose input size, same temporal classifier. Four timing clips × two repetitions per backend; bars show mean ± sample SD.

Paired throughput

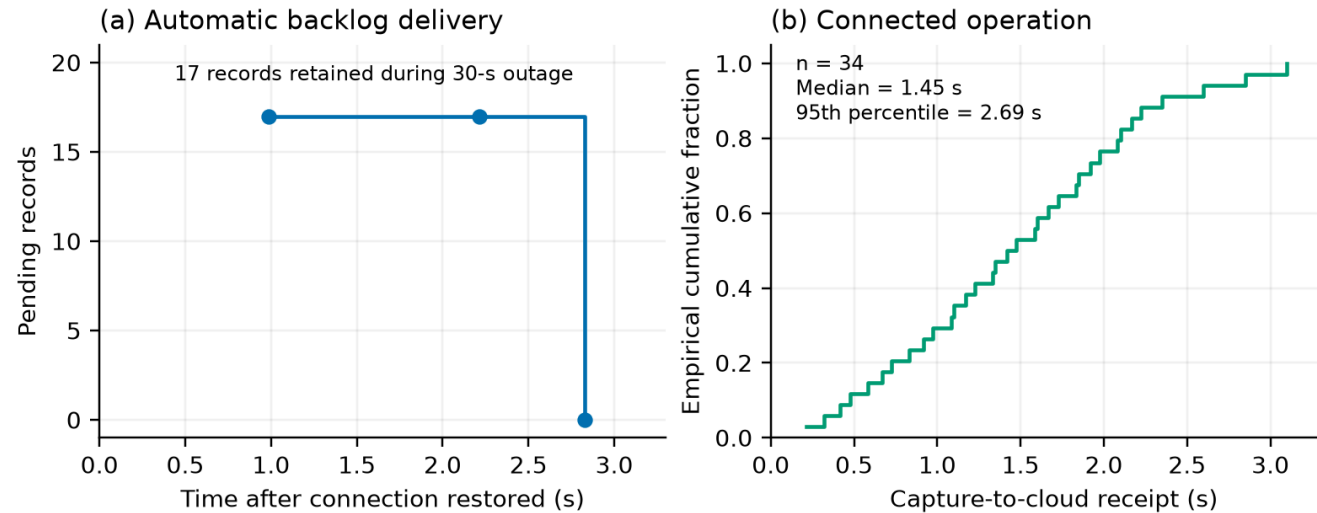
20.56 ± 0.39 fps with Hailo, versus **1.73 ± 0.02 fps** with CPU ONNX. The complete pipeline includes decoding, pose extraction, classification and record writes.

Forty synthetic clips

39/40 valid outputs; 20 true positives, 11 true negatives, 8 false positives and 1 missing output. Accuracy among classified clips: **79.49%**. Clip F1: **0.8333**.

Only 14/20 positive clips produce a positive window ending within the annotated fall interval. Clip-level recall is not the same as detection during the fall.

Records survive interruptions



Queue markers are sampled observations. Connected latency starts at the final captured frame of a classification window, not at fall onset.

85

edge/cloud records matched

100 / 100

committed writes recovered

57.85 °C

peak in the 10-minute replay

A 30-second upload-path outage leaves 17 pending records. The queue is first observed empty 2.83 seconds after recovery. Replaying 34 records after a cloud restart creates no duplicates.

Show the device working independently

01 Connect a display

Micro-HDMI from the Pi to a monitor or projector. Open the local dashboard.

02 Show valid observations

Frame the whole body; show skeletons, current predictions and saved records.

03 Disconnect the network

Turn off Wi-Fi and unplug Ethernet. Continue inference and show pending records.

04 Reconnect

Show the queue drain and the cloud history retain original event times.

DEVICE DISPLAY

Local and cloud workspaces

Overview: preview and fresh results

History: filters, pagination and CSV

Diagnostics: resources and delivery

Invalid or expired probability is cleared

Cloud history remains during edge loss

The local page is served by the Pi. This presentation is a static evidence viewer and does not connect to the device or cloud backend.

Before the demonstration, check full-body framing and increasing classification counts. Public or synthetic local videos provide a clearly labeled fallback demonstration.

Workspace update verified 7 October 2026: 3,304 classified windows retained; queue empty at the check. This is an operational snapshot, not a new accuracy sample.

What the project establishes

Delivered

- A quality-aware bidirectional temporal classifier, with baseline, ablation and robustness evaluation.
- Independent Pi pose extraction and fall classification, accelerated by Hailo.
- Local records, recovery after interruption, and verified cloud delivery.

MaskedBiMamba retains more window F1 under frame removal than TCNTE, and achieves the highest precision at the common threshold.

Next improvements

- Calibrate confirmation rules on continuous daily-activity recordings to reduce false alerts.
- Extend camera and cross-dataset evaluation; Le2i specificity remains the main transfer challenge.
- Evaluate longer operation and energy use across realistic observation conditions.

TE leads clean F1. Model selection here prioritizes precision, tolerance to missing frames and a complete edge-to-cloud implementation.

Thank you.

[Explore every aggregate result →](#)